Importing a Life: Duplicate and Near-Duplicate Detection Across Personal Information Sources

Paola Rinaldi, Farhan Uddin

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Abstract

Personal information tools increasingly offer to import everything a user has: email, files, notes, chats and references. Imports produce duplicates. We collected consented exports from 14 participants across 9 source types and annotated duplicate and near-duplicate pairs. 18% of imported items had at least one duplicate. Exact hashing found 41% of duplicate pairs, title-and-size matching 73%, and content shingling 92%, but shingling also falsely merged successive versions of drafts that participants wanted kept apart. We propose a two-stage policy: merge exact duplicates silently, group near-duplicates for the user to confirm.

1 Introduction

The same PDF arrives through a reference manager, the downloads folder and an email attachment. The same draft exists as "report", "report v2" and "Copy of report". A tool that imports all of these without care shows the user three of everything.

2 Data

14 participants (graduate students and knowledge workers) shared exports from up to 9 sources: email, cloud drive, local files, reference manager, note app, chat assistant, calendar, code repositories and messaging. We annotated 6,410 candidate pairs.

Duplicate kind	Pairs	Example
Exact byte copy	512	same PDF in Downloads and the reference manager
Same work, different file	388	preprint vs published version
Draft versions	446	report.docx vs report v2.docx
Same content, different format	207	note exported as Markdown and HTML

3 Detection

Method	Recall (all dup pairs)	False merges of draft versions
Exact hash	41%	0%
Title + size	73%	12%
Content shingling (Jaccard >= 0.8)	92%	61%

Participants were clear that preprint and published versions of a paper are "the same paper" for reading, but drafts of their own writing are not duplicates: each version marks a decision.

4 Policy

Merge exact duplicates silently and keep every location as provenance. Group same-work items (for example by DOI or normalized title) under one entry. Show near-duplicate drafts as a version stack the user can expand, never merged.

5 Limitations

Fourteen participants; English-language content; exports taken at one point in time.

References

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